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ChE 210 Exam #1 11 October 2019

(1) (25 pts.) Consider a mixture of gasses consisting of three species, namely, A, B, and C. It is known that the mixture is made up of 40 mol% A, 18.75 wt % of B, 20 mol% of C. The molecular weight of A is 40kg/kmole and of C is 50 kg.kmole.

(a) Calculate the molecular weight of B in kg/kmol.

mix

3 gasses

A 40mol% A

B 18.75wt% → 40mol%

+ C 20mol% C

100 mol

Total mol 100 mol in mix

MW: A = 40kg/kmol

C = 50kg/kmol

$40\text{mol A} \cdot \frac{1 \text{ kmol A}}{10^3 \text{ mol A}} \cdot \frac{40\text{kg A}}{\text{kmol}} = 1.6\text{kg A}$

$20\text{mol C} \cdot \frac{1\text{kmol C}}{10^3\text{mol C}} \cdot \frac{50\text{kg C}}{1\text{kmol C}} = 1\text{ kg C}$

0.6 kg B

Total

$\frac{x}{1.6+1+x} = 0.1875$

$\frac{x}{2.6+x} = 0.1875(2.6+x)$

$x = 0.4875 + 0.1875x$

$0.8125x = 0.4875$

$x = 0.6\text{ kg B}$

$0.6\text{kg B} \cdot \frac{1\text{ kmol}}{x\text{ kg B}} = 0.04\text{ kmol}$

$\frac{0.6}{x} = 0.04$

$M_B =$ 15 kg/kmol

1	25
2	39
3	23
T	87

(b) Calculate the mole average molecular weight of the mixture kg/kmole.

1.6kg A/mol

1 kg C

+ 0.6 kg B

$\frac{3.2\text{kg mix}}{100\text{mol mix}} \cdot \frac{10^3\text{ mol mix}}{1\text{ kmol mix}} = 32$ units

25

$\bar{M} =$ 32 kg/kmol

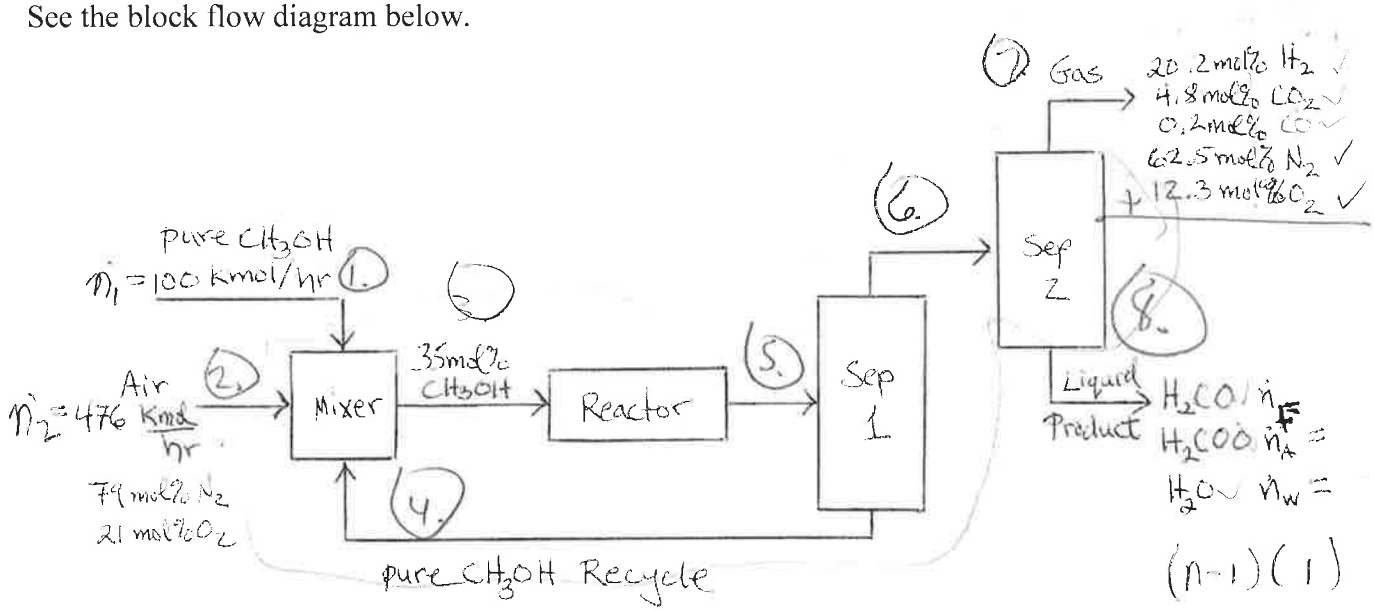
(2) (50 pts.) Formaldehyde (H_2CO) is produced by oxidation of methanol (CH_3OH). As shown, four reactions take place in the reactor. Pure methanol (CH_3OH) at 100 kmol/hr along with 476 kmol/hr of air (21 mol% O_2 and 79 mol% N_2) and a recycle stream of pure methanol are mixed and then enter the reactor. The reactor inlet stream is 35 mol% CH_3OH. The reactor exit stream goes to a separator with one exit stream of pure methanol that is recycled to the mixer and another stream that goes to a second separator that has an exit stream of gas with 20.2 mol% H_2, 4.8 mol % CO_2, 0.2 mol % CO, 62.5 mol% N_2, and 12.3 mol % O_2 and a liquid product stream containing formaldehyde (H_2CO), formic acid (H_2COO), and water (H_2O). See the block flow diagram below.

$$CH_3OH + \frac{1}{2}O_2 \rightarrow HCHO + H_2O \qquad \text{(R1)}$$

$$CH_3OH + O_2 \rightarrow HCOOH + H_2O \qquad \text{(R2)}$$

$$CH_3OH + O_2 \rightarrow CO + 2H_2O \qquad \text{(R3)}$$

$$CH_3OH + \frac{3}{2}O_2 \rightarrow CO_2 + 2H_2O \qquad \text{(R4)}$$



(a) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. Fill in the table below giving a brief explanation for your entries. Compute the DOF. (10 pts.)

	Number	Explanation
4a. Stream Variables	~~11~~	1 in (1), 2 in (2), 5 in (7), 3 in (8)
4b. System Variable	4	4 rxns
Total Independent Variables	15	
5a. Specified Flows	2	n_1 and n_2
5b. Specified Stream Compositions	5+1	n_2, 2-1=1, (7) Sep (5-1)(2-1)=4 (8) 0
5c. Specified System Performances	0	None
5d. Material Balance Equations	~~9~~	CH_3OH, N_2, O_2, H_2, CO_2, CO, H_2COO, H_2CO, H_2O
Total Independent Equations	16	7
DOF	15-16=-1	over specified

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PROBLEM 2 CONTINUED in - out

(b) Calculate the product stream molar flow rates in kmol/hr for the formaldehyde (H_2CO) $\dot{n}_F$, formic acid (H_2COO) $\dot{n}_A$, and water (H_2O) $\dot{n}_W$ in kmol/hr. *Hint: Consider an atomic balance.* (30 pts.)

$$n_1 + n_2 = \dot{n}_7 + \dot{n}_8$$
$$100 + 476 = n_7 + n_8$$
$$576 = n_7 + \dot{n}_8$$

$$H: 0 = 4(100) - 2(0.202\dot{n}_7) - 2n_F - 2n_A - 2\dot{n}_W$$
$$C: 0 = 1(100) - 1(0.048\dot{n}_7) - 1(0.02\dot{n}_7) - n_F - n_A - \dot{n}_W$$
$$O: 0 = 1(100) + 2[0.21(476)] - 2(0.048\dot{n}_7) - 0.2\dot{n}_7 - 2(0.123\dot{n}_7)$$
$$- \dot{n}_F - 2\dot{n}_A - \dot{n}_W$$

$$N: 0 = 2(0.79(476)) - 2(0.625\dot{n}_7)$$ $n_8 = -25.664$ ✓ YES! makes sense since werspecifical

$$-752.08 = -1.25\dot{n}_7 \quad \dot{n}_7 = 601.664 \text{ kmol/hr}$$

$$-156.93 = -2n_F - 2n_A - 2\dot{n}_W$$

$$n_F = -n_A - \dot{n}_W + 59.08$$

$$\dot{n}_F = 78.465 + n_A + \dot{n}_W$$

27

$\dot{n}_F$ = 150 kmol H_2CO /min, $\dot{n}_A$ = 250 kmol H_2COO /min, $\dot{n}_W$ = 400 kmol H_2O/min

(c) Calculate the fractional conversion of methanol for the reactor (the single-pass conversion). (10 pts.)

conversion 100% converted since none left on output Stream 5

$$\text{consumed} = \frac{\text{consumed}}{\text{Beginning } CH_3OH} \times 100\%$$

5

39

$f_{C,methanol}$ = ________ Selectivity = ________

(25 pts.) The Ostwald Process produces nitric acid (HNO_3) by the oxidation of ammonia (NH_3). The process consists of three reactions:
R1: Ammonia reacts with oxygen (O_2) to form nitric oxide (NO) and water (H_2O).
R2: NO reacts with O_2 to form nitrogen dioxide (NO_2).
R3: NO_2 is bubbled through water to produce HNO_3 and NO.

(a) Write a balance chemical reaction equation for each of these three reaction. Make all of the stoichiometric coefficients the smallest whole numbers possible.

$4NH_3 + 5O_2 \longrightarrow 4NO + 6H_2O$ ✓

$2NO + O_2 \longrightarrow 2NO_2$ ✓

$3NO_2 + H_2O \longrightarrow 2HNO_3 + NO$ ✓

(b) Use a generation-consumption analysis to synthesize a reaction pathway to make nitric acid from ammonia and oxygen based on the three reaction in part (a) with no net generation or consumption of NO or NO_2. Water is allowed as a byproduct.

	R1	R2	R3	Overall Rxn
NH_3	−4			−4
O_2	−5	−1 (3)		−8
NO	+4	−2 (3)	+1 (2)	0
NO_2		+2 (3)	−3 (2)	0
H_2O	+6		−1	5
HNO_3			+2	2

Write the net chemical equation for the process.

$4NH_3 + 8O_2 \longrightarrow 5H_2O + 2HNO_3$ −2

(c) Calculate the grams of NH_3 and grams of O_2 required to make 100 grams of nitric acid. Atomic Weights: H = 1 g/mol, N = 14 g/mol, and O = 16 g/mol.

$100\,g\,HNO_3 \cdot \frac{1\,mol\,HNO_3}{63\,g\,HNO_3} \cdot \frac{4\,mol\,NH_3}{2\,mol\,HNO_3} \cdot \frac{17\,g\,NH_3}{1\,mol\,NH_3} = 53.97\,g\,NH_3$

$\frac{8\,mol\,O_2}{2\,mol\,HNO_3} \cdot \frac{32\,g\,O_2}{1\,mol\,O_2}$

23

Mass of O_2 = ~~203.17~~ g OK −2

Mass of NH_3 = ~~53.97~~ g ok −2